Lecture 3 // Sept 1 // Examples & Hodge Theory

Recall

A connection is a map $\nabla : \Gamma(E) \to \Gamma(T^*M \oplus E)$. Locally, connections take the form $\nabla = d + A$ where A is a connection 1-form matrix. Under change of coordinates, A transforms strangely, not by conjugation:

$$A \mapsto g^{-1}Ag + g^{-1}dg$$

where g is the bundle transition function. The curvature of a connection is $F_A := dA + A \wedge A$, and by Chern–Weil theory, the Chern classes are

$$\det\left(I - \frac{t}{2\pi}F_A\right).$$

Today, we'll explicitly compute one connection & its curvature to illustrate them in a toy example. This example will also generalize to help us understand the Yang–Mills moduli space.

Example: Connections on $\mathbb{CP}^1 = S^2$

Recall that $\mathbb{CP}^1 = \{[z : w] \mid z, w \in \mathbb{C}^2\}$ can be covered with 2 charts, the z chart with all points $[z : w]$ but $[0 : 1]$, and the w chart. The transition function is $w = \frac{1}{z}$.

Now, consider the $\mathbb{C}$-line bundle over $\mathbb{CP}^1$ defined by the bundle transition function $g(z) = z^n$. This is the degree-n $\mathbb{C}$-line bundle where $\mathbb{C}$-fibers get glued by z^n, viewing this as a 1×1 matrix over $\mathbb{C}$.

Fact 1

These are all the $\mathbb{C}$ line bundles over $\mathbb{CP}^1$.

What do connections on the degree-n line bundle look like? To define a connection, we must write down the connection's 1-form matrix A. Since the bundle is a line bundle, A is a 1×1 matrix over $\mathbb{C}$, i.e., a function $\mathbb{C} \to \mathbb{C}$.

Since A is purely local, we must define it in local coordinates. We'll consider the z chart first.

Proposition 2

$A := -n\dfrac{\bar{z}\,dz}{1 + |z|^2}$ defines a connection on the degree n $\mathbb{C}$-line bundle over $\mathbb{CP}^1$.

Proof. To check that A actually defines a connection, we must do the following:

1. look at A in the other chart
2. show that A extends smoothly over the point at ∞.

Each step requires its own coordinate change:

1. $A \rightsquigarrow g^{-1}Ag + g^{-1}dg$, the bundle change of coordinates

2. $w = \frac{1}{z}$, the manifold change of coordinates

1) Recall that the bundle transition function is $g(z) = z^n$.

Thus

$$\begin{aligned} g^{-1}Ag + g^{-1}dg &= z^{-n}\left(-n\frac{\bar{z}\,dz}{1+|z|^2}\right)z^n + z^{-n}d(z^n) \\ &= -n\frac{\bar{z}\,dz}{1+|z|^2} + \frac{nz^{n-1}}{z^n} \\ &= -n\frac{\bar{z}\,dz}{1+|z|^2} + \frac{n}{z} \end{aligned}$$

2) We now apply the manifold change of coordinates: $w = \frac{1}{z}$

$$dz = d\left(\frac{1}{w}\right) = \frac{-dw}{w^2}$$

So $-n\frac{\bar{z}\,dz}{1+|z|^2} + \frac{n}{z}$ becomes

$$\begin{aligned} \frac{-n\left(\frac{1}{w}\left(\frac{-1}{w^2}dw\right)\right)}{1+\left|\frac{1}{w}\right|^2} + nw\left(\frac{-1}{w^2}\right)dw &= -n\frac{dw}{w^3\left(1+\left|\frac{1}{w}\right|^2\right)} - n\frac{dw}{w} \\ &= \frac{-n\left(\frac{1}{w}\left(\frac{-1}{w^2}dw\right)\right)}{1+\left|\frac{1}{w}\right|^2} + nw\left(\frac{-1}{w^2}\right)dw \\ &= -n\frac{dw}{w^3\left(1+\left|\frac{1}{w}\right|^2\right)} - n\frac{dw}{w} \\ &= n\frac{dw}{w(1+|w|^2)} - n\frac{dw}{w} \\ &= n\left(\frac{1}{w(1+|w|^2)} - \frac{1}{w}\right)dw \\ &= n\left(\frac{1}{w(1+|w|^2)} - \frac{(1+|w|^2)}{w(1+|w|^2)}\right)dw \\ &= n\left(\frac{-|w|^2}{w(1+|w|^2)}\right)dw \\ &= \frac{-n\bar{w}\,dw}{1+|w|^2} \end{aligned}$$

this is a valid 1-form on the w-chart, and is defined at $z = \infty \leftrightarrow w = 0$.

Thus A defines a valid global 1-form. □

Note that A itself does not extend as a smooth 1-form on $\mathbb{CP}^1$, since $n\frac{dw}{w(1+|w|^2)}$ is singular at $w = 0 \leftrightarrow z = \infty$. It is precisely the addition of the $g^{-1}dg = -n\frac{dw}{w}$ term from the bundle change of coordinates which resolves the issue. So A can be defined globally because of the "strange" bundle change-of-coordinates.

We'll now compute the curvature of this 1-form: $F_A = dA + A \wedge A$. Since $d^2 = 0$, $A \wedge A = 0$, we need

only to compute dA.

$$\begin{aligned} dA = d\left(-n\frac{\bar{z}\,dz}{1+|z|^2}\right) &= -n\left(\frac{(1+|z|^2)d\bar{z} - \bar{z}(zd\bar{z}+\bar{z}dz)}{(1+|z|^2)^2}\right)\wedge dz \\ &= -n\frac{d\bar{z}\wedge dz}{(1+|z|^2)^2} \qquad d\bar{z} \\ &= dx - i\,dy, \quad dz = dx + i\,dy \\ &= -2ni\frac{dx\wedge dy}{(1+x^2+y^2)^2} \end{aligned}$$

Thus $F_A = -2ni\dfrac{dx\wedge dy}{(1+x^2+y^2)^2}$.

Using the formula for the Chern classes from Chern-Weil theory, we can compute the first Chern class; note that this is the expression of c_1 on the z-chart.

$$\Rightarrow c_1 = \frac{-1}{2\pi i}\left(-2ni\frac{dx\wedge dy}{(1+x^2+y^2)^2}\right) = \frac{n}{\pi}\frac{dx\wedge dy}{(1+x^2+y^2)^2}$$

Integrating over $\mathbb{CP}^1$, we get that

$$\begin{aligned} \int_{\mathbb{CP}^1} c_1 &= \frac{n}{\pi}\left(\int_{\mathbb{R}^2}\frac{dx\wedge dy}{(1+x^2+y^2)^2}\right) \\ &= \frac{n}{\pi}\int_0^{2\pi}\int_0^{\infty}\frac{r}{(1+r^2)^2}\,dr\,d\theta \\ &= \frac{n}{\pi}\int_0^{2\pi}\frac{1}{2}\,d\theta \\ &= n \end{aligned}$$

The previous computation leaves us with some questions.

1. If the final integral was taken over $\mathbb{R}^2$, where did $\mathbb{CP}^1$ come in to play? Why can't we choose another A defined only on $\mathbb{R}^2$?

2. In this example,
$$\int_{\mathbb{CP}^1} c_1 = \int_{\mathbb{CP}^2} F_A = \int_{\mathbb{CP}^2} dA = \int_{\text{large disk}} dA = \int_{\text{large } S^1} A.$$
Could we instead choose A to be compactly supported, so $\int = 0$? Or even choose $A = 0$?

The answer to both is a resounding no. The point here is that not every choice of A defines a connection on a $U(1)$ bundle over $\mathbb{CP}^1$. In this example, even though A is integrated over $\mathbb{R}^2$, we already checked that it extends, so the fact that A is defined globally comes first. A "knows" what bundle it lives on by "remembering" the singular behavior at $z=\infty$, since it must match the $\frac{n\,dw}{w}$ term to allow for cancellation.

If $A = 0$ or is compactly supported, $\dfrac{dw}{w(1+|w|^2)} = 0$, so $n\frac{dw}{w(1+|w|^2)} - n\frac{dw}{w} = -n\frac{dw}{w}$. For this term to be nonsingular, n must be 0, so the line bundle must be trivial. The upshot here is that the curvature integral determines the bundle via the behaviour near ∞.

A Review of Hodge Thery

Let (M, g) be an oriented Riemannian manifold. Recall that on each tangent space T_pM, we get an inner product pairing $\langle \cdot, \cdot \rangle_p$ which smoothly varies as p varies in M. This gives an inner product on each cotangent space T_p^*M which then induces an inner product on $\Lambda^k T^*M$ for each k as follows. Let $\{e_1, \dots, e_n\}$ be an oriented orthonormal basis of T_pM. Dualizing, we get a basis $\{e_1^*, \dots, e_n^*\}$ of T_p^*M which we may declare to be oriented and orthonormal; this declaration induces the inner product structure. For each fixed k, $\Lambda^k T^*M$ is $\binom{n}{k}$-dimensional. For each choice of k numbers between 1 and n, $\{i_1, \dots, i_k\} \subset \{1, \dots, n\}$, we get a basis vector of the form $e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$. Again, we declare this basis to be oriented and orthonormal, which induces the inner product structure on $\Lambda^k T^*M$. Of course, one must check that the resulting inner product doesn't depend on the original choice of oriented orthonormal basis for T_pM.

By abuse of notation, all of these inner products will be denoted by $\langle -, - \rangle$.

Now, let $n = \dim M$. On each $\Lambda^n T_p^*M$, there exists a unique oriented n-wedge with norm 1 via the above identification. This n-wedge is the volume form $e_1^* \wedge \cdots \wedge e_n^* =: d\mathrm{vol}_g$. Also that since $\Lambda^k T^*M$ has $\binom{n}{k}$ basis vectors, $\dim(\Lambda^k T^*M) = \dim(\Lambda^{n-k} T^*M)$; this is also true for differential forms. To exploit this symmetry, we may define the Hodge star.

Definition 3

The Hodge star is a map $\star : \Omega^k \to \Omega^{n-k}$ defined pointwise by (after fixing an orthonormal oriented basis $\{e_1, \dots, e_n\}$ of T_pM): for each $I \subset \{1, \dots, n\}$ with $|I| = k$, define $e_I^* := e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$.

For each basis vector e_I^*, the Hodge star sends e_I^* to $\star e_I^*$ via

$$\star e_I^* := \pm \underbrace{e_{j_1}^* \wedge \cdots \wedge e_{j_{n-k}}^*}_{e_J^*} \quad \text{where } J = \{1, \dots, n\} \setminus I.$$

The $\pm$ sign is chosen such that $e_I^* \wedge e_J^* = e_1^* \wedge \cdots \wedge e_n^*$, so $\star$ gives the complementary wedge of forms. To define $\star$ on non-basis vectors, we extend linearly.

Note that this definition is independent of the choice of orthonormal basis, but it does depend on the choice of metric. Furthermore, $\star$ is almost an involution... but not quite. To illustrate this behavior, consider $e_I = e_1 \wedge e_3$ and $e_J = -e_2 \wedge e_4$. Then

$$\begin{aligned} e_I \wedge e_J &= e_1 \wedge e_3 \wedge -e_2 \wedge e_4 \\ &= e_1 \wedge e_2 \wedge e_3 \wedge e_4 \\ &= d\mathrm{vol}_g \end{aligned}$$

so $\star(e_I) = e_J$. Similarly,

$$\begin{aligned} e_J \wedge e_I &= -e_2 \wedge e_4 \wedge e_1 \wedge e_3 \\ &= -e_2 \wedge -e_1 \wedge e_4 \wedge e_3 \\ &= e_1 \wedge -e_2 \wedge e_4 \wedge e_3 \\ &= e_1 \wedge -e_2 \wedge -e_3 \wedge e_4 \\ &= e_1 \wedge e_2 \wedge e_3 \wedge e_4 \\ &= d\text{vol}_g \end{aligned}$$

Thus $\star(e_J) = e_I$, so in this case, $\star^2(e_I) = e_I$. However, this is a direct result of the number of swaps we had to perform to correctly order the basis vectors. In general, the total number of swaps depends on n and k: $(\star)^2 = (-1)^{k(n-k)}$.

Defining the Hodge star buys us two things. First, we can get an inner product structure on Ω^k from that on $\Lambda^k T^*M$, which was induced by the metric.

Definition 4
For $\alpha, \beta \in \Omega^k$, define

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge \star\beta \qquad \text{(since } \alpha \wedge \star\beta \text{is an n-form.)}$$

If α is an orthonormal basis element, $\alpha \wedge \star\alpha = d\text{vol}_g$, so

$$\langle \alpha, \alpha \rangle = \int_M |\alpha|^2 \, d\text{vol}_g.$$

Second, we get a formal adjoint of the exterior derivative. Recall that the exterior derivative is a map $d : \Omega^k \to \Omega^{k+1}$. A formal adjoint of d then must be a map $d^* : \Omega^k \to \Omega^{k-1}$.

Definition 5
The "codifferential" $d^* : \Omega^k \to \Omega^{k-1}$ is defined as the composition $d^* := (-1)^k \star d \star$:

$$d^* : \Omega^k \xrightarrow{\star} \Omega^{n-k} \xrightarrow{d} \Omega^{n-k+1} \xrightarrow{\star} \Omega^{k-1}.$$

Furthermore, $(d^*)^2 = 0$.

Lemma 6
d^* is the formal adjoint of d: $\langle d\alpha, \beta \rangle = \langle \alpha, d^*\beta \rangle$.

Proof. By Stokes' theorem and the product rule,

$$0 = \int_M d(\alpha \wedge \star\beta) = \int_M d\alpha \wedge \star\beta \pm \int_M \alpha \wedge d(\star\beta)$$

$$= \langle d\alpha, \beta \rangle \pm \int_M \alpha \wedge \star d^* \beta = \langle d\alpha, \beta \rangle \pm \langle \alpha, d^* \beta \rangle$$
$$\Rightarrow \langle d\alpha, \beta \rangle = \mp \langle \alpha, d^* \beta \rangle$$ □

We now have the tools to discuss the main point of Hodge theory (for our purposes).

Definition 7
The Hodge Laplacian is a map $\Delta : \Omega^k \to \Omega^k$ defined as

$$\Delta = (d + d^*)^2 = dd^* + d^* d.$$

k-forms $\alpha \in \Omega^k$ are called harmonic if they are in the kernel of the Hodge Laplacian, eg, $\Delta\alpha = 0$.

Lemma 8
α is harmonic iff $d\alpha = 0$ (closed) and $d^*\alpha = 0$ (co-closed).

Proof.

($\Leftarrow$) This obvious by the definition.

($\Rightarrow$) If $\Delta\alpha = 0$, $\langle \Delta\alpha, \alpha \rangle = 0$. Since $\Delta\alpha = dd^*\alpha + d^* d\alpha$,

$$0 = \langle \Delta\alpha, \alpha \rangle = \langle dd^*\alpha + d^* d\alpha, \alpha \rangle = \langle dd^*\alpha, \alpha \rangle + \langle d^* d\alpha, \alpha \rangle$$
$$= \langle d^*\alpha, d^*\alpha \rangle + \langle d\alpha, d\alpha \rangle = \langle (d + d^*)\alpha, (d + d^*)\alpha \rangle$$
$$\Rightarrow (d + d^*)\alpha = 0 \Rightarrow d\alpha = d^*\alpha = 0.$$ □

If α is harmonic, it is closed, so α defines a cohomology class $[\alpha] \in H^k_{dR}$. Furthermore, in local coordinates, $d^*d + dd^*$ is a 2nd order differential operator, since it acts like a 2nd derivative on the coefficients of basis vectors. We can interpret harmonicity in this context as cohomology classes which are norm minimizing. This gives the following lemma.

Lemma 9
α is harmonic iff α minimizes norm in its de Rham cohomology class.

Proof.

($\Rightarrow$) Let $\alpha \in H^k_{dR}(M; \mathbb{R})$. Every other element in this class will differ from α by an exact form, i.e., some $d\beta$ for some $\beta \in \Omega^{k-1}$. We will compute the norm of $\alpha + d\beta$.

$$\langle \alpha + d\beta, \alpha + d\beta \rangle = \langle \alpha, \alpha \rangle + \langle d\beta, d\beta \rangle + 2\langle \alpha, d\beta \rangle$$

We know that $\langle d\beta, d\beta \rangle \geq 0$, and $\langle \alpha, d\beta \rangle = \langle d^*\alpha, \beta \rangle = 0$ since α is harmonic.

Thus $\langle \alpha, \alpha \rangle \leq \langle \alpha + d\beta, \alpha + d\beta \rangle$.

($\Leftarrow$) Assume α is norm minimizing. For all $\beta \in \Omega^{k-1}$, consider the ray $\langle \alpha + t\, d\beta, \alpha + t\, d\beta \rangle_{t\in[0,1]}$. I.e., we move α around in its cohomology class along some ray. Since α is norm minimizing, when $t = 0$, the derivative vanishes.

$$0 = \left.\frac{d}{dt}\right|_{t=0} \langle \alpha + t\, d\beta, \alpha + t\, d\beta \rangle = 2\langle \alpha, d\beta \rangle = 2\langle d^*\alpha, \beta \rangle$$

$$\Rightarrow d^*\alpha = 0$$

It remains to show $d\alpha = 0$ and that this minimizer is unique; the proof is left as an exercise. $\square$

Theorem 10 (Hodge Theorem)
Define $\mathcal{H}^k$ to be the space of harmonic k-forms. There exists an isomorphism $\mathcal{H}^k \cong H^k_{dR}$ sending $\alpha \in H^k_{dR}$ to the norm minimizer in $[\alpha]$.